

6(a)	energy is dissipated in the internal resistance	B1
6(b)	$E = V + Ir$	B1
6(c)(i)	(graph shows) maximum value of potential difference is 2.8 (V) or (graph shows) when current/ I (from battery) is zero, V is 2.8 (V) / E	B1
6(c)(ii)	$r = (-)\text{gradient}$ or $r = (E - V) / I$ or substituted values from the graph for E , V and I	C1
	$r = 1.4 \Omega$	A1
6(d)(i)	$R = 2.1 / 0.50$ $= 4.2 \Omega$	A1
6(d)(ii)	number = $0.50 / 1.60 \times 10^{-19}$ $= 3.1 \times 10^{18}$	A1
6(d)(iii)	energy = EIt or $P = EI$ and $P = W / t$	C1
	$(9.2 - 1.6) \times 10^3 = 2.8 \times 0.50 \times t$	C1
	$t = 5.4 \times 10^3 \text{ s}$	A1